CHAPTER

05

Oscillations, Waves and Optics

An **oscillation** is a disturbance that **repeats in time** (a pendulum swinging, a mass bouncing on a spring). A **wave** is a disturbance that repeats in time **and** in space, and that **carries energy and information from place to place** without carrying matter. Sound, light, radio, water ripples and earthquakes are all waves, and they all obey the same mathematics. The chapter ends with **optics**: how mirrors and lenses form images.

LESSONS IN THIS CHAPTER

5.1 Simple Harmonic Motion

5.2 The Simple Pendulum

5.3 Waves and Their Characteristics

5.4 Resonance and the Doppler Effect

5.5 Image Formation by Mirrors and Thin Lenses

Freshman A+ · General Physics (Phys 1011)

CHAPTER 5 · LESSON 5.1

Simple Harmonic Motion

AFTER THIS LESSON YOU CAN

- Distinguish periodic, oscillatory and simple harmonic motion
- Define period, frequency, amplitude and angular frequency
- Write and use x(t), v(t) and a(t) for SHM
- Analyse the mass–spring system and the energy of SHM

5.1.1 Periodic and oscillatory motion

D KEY DEFINITIONS

- **Periodic motion:** any motion that repeats itself in equal intervals of time (the Earth round the Sun, a clock hand).
- **Oscillatory (vibratory) motion:** periodic motion **to and fro along the same path** about an **equilibrium position**, caused by a **restoring force** that grows with the displacement.
- **Simple harmonic motion (SHM):** oscillation in which the restoring force obeys **Hooke's law**, $F = -kx$, so the **acceleration is proportional to the displacement and opposite in direction**: $a = -\omega^2 x$.
  Every oscillation is periodic, but not every periodic motion is an oscillation. A ball tossed back and forth between two people repeats its motion but is **not SHM** (the force is not $-kx$).

Mechanical oscillations need two things: **inertia** (to overshoot the equilibrium) and a **restoring force** (to pull back). In a **non-mechanical** oscillator, nothing moves but a physical property varies periodically (e.g. the current in an LC circuit).

F DESCRIBING AN OSCILLATION

Quantity	Meaning	Formula / unit
Amplitude A	maximum displacement from equilibrium	m
Period T	time for **one complete cycle**	s
Frequency f	number of cycles per second	$f = 1/T$, Hz
Angular frequency ω		$\omega = 2\pi f = 2\pi/T$, rad/s

✎ WORKED EXAMPLE 1: THE HEARTBEAT (MODULE)

A heart beats 75 times per minute. $f = 75/60 = 1.25$ Hz; $T = 1/f = 0.80$ s.

5.1.2 The mass–spring system

A block of mass m on a spring of constant k (frictionless surface). Newton's second law with Hooke's law:

$$ma = -kx \quad \Longrightarrow \quad a = -\frac{k}{m}x = -\omega^2 x, \qquad \omega = \sqrt{\frac{k}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}, \qquad f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

CHARACTERISTICS OF SHM

1. The amplitude is **constant** (no energy loss: undamped).
2. The **period and frequency do not depend on the amplitude**.
3. The motion is described by a **single sine or cosine** function.

Conditions for SHM: a stable equilibrium position; no dissipation of energy; acceleration proportional to displacement and opposite in direction. (A constant velocity is **not** a condition; in SHM the velocity keeps changing.)

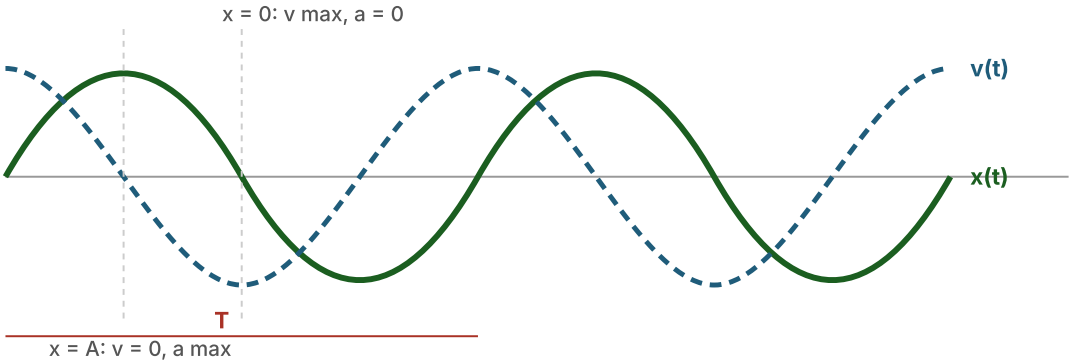
5.1.3 Displacement, velocity and acceleration

$$x = A\sin(\omega t + \phi) \qquad v = \frac{dx}{dt} = A\omega\cos(\omega t + \phi) \qquad a = \frac{dv}{dt} = -A\omega^2\sin(\omega t + \phi) = -\omega^2 x$$

(ϕ is the **phase constant**, fixed by where the motion starts. A cosine form works the same way.)

MAXIMUM VALUES AND WHERE THEY OCCUR

	At equilibrium ($x = 0$)	At the extremes ($x = \pm A$)
Speed	**maximum**: $v_{\max} = \omega A$	zero
Acceleration	zero	**maximum**: $a_{\max} = \omega^2 A$
Restoring force	zero	maximum: $F_{\max} = kA = m\omega^2 A$
Speed at any position: $v = \omega\sqrt{A^2 - x^2}$.		



Displacement (solid) and velocity (dashed) in SHM: velocity is a quarter cycle ahead, and acceleration is always opposite to x

WORKED EXAMPLE 2: READING AN SHM EQUATION (SERWAY / MODULE)

$x = (4.00\ \text{m})\cos\left(\pi t + \frac{\pi}{4}\right)$. **Find (a)** A, f, T; **(b)** x, v, a **at** $t = 1.00$ **s; (c)** $v_{\max}$ **and** $a_{\max}$.

(a) $A = 4.00$ m; $\omega = \pi$ rad/s → $f = \omega/2\pi = 0.500$ Hz; $T = 2.00$ s.

(b) Phase at $t = 1$: $\pi + \pi/4 = 5\pi/4$.

$x = 4.00\cos(5\pi/4) = -2.83$ m; $v = -4.00\pi\sin(5\pi/4) = +8.89$ m/s; $a = -4.00\pi^2\cos(5\pi/4) = +27.9$ m/s².

(c) $v_{\max} = A\omega = 4\pi = 12.6$ m/s; $a_{\max} = A\omega^2 = 4\pi^2 = 39.5$ m/s².

5.1.4 Energy in SHM

$$U = \tfrac{1}{2}kx^2 \qquad K = \tfrac{1}{2}mv^2 \qquad E = K + U = \tfrac{1}{2}kA^2 = \tfrac{1}{2}mv_{\max}^2 = \text{constant}$$

Energy flows back and forth between kinetic and potential: all potential at the extremes, all kinetic at equilibrium. **The total energy is proportional to the square of the amplitude**: doubling A quadruples E.

SHM TRAPS

- **Velocity and acceleration** can point the same way (when moving toward equilibrium); **position and acceleration never can** ($a = -\omega^2 x$).
- The period of a spring system depends on m and k, **not** on A or g.
- $v_{\max}$ is at the **centre**, $a_{\max}$ at the **ends**.
- "Frequency" in Hz is $\omega/2\pi$; don't confuse f and ω.

Exam Practice

12 QUESTIONS · LESSON 5.1

1 Wolaita Sodo University · Final Exam 2014

Which of the following is NOT a necessary condition for simple harmonic oscillation?

A) There must be a stable equilibrium position
B) There must be no dissipation of energy
C) The acceleration is proportional to the displacement and opposite in direction
D) There must be a constant velocity

2 Wolaita Sodo University · Final Exam 2014

Which of the following is true for undamped SHM?

A) The amplitude increases with time
B) The acceleration is proportional to the displacement and opposite in direction
C) The period depends on the amplitude
D) There is continuous dissipation of mechanical energy

3 Wolaita Sodo University · Final Exam 2014

The number of oscillations performed per unit time is called:

A) Time period
B) Amplitude
C) Frequency
D) Wavelength

4 Arba Minch University · Final Exam

A particle in SHM with amplitude 5 cm has a maximum speed of $20\pi^2$ cm/s. Its frequency is:

A) 3π Hz
B) 4π Hz
C) 2π Hz
D) $2\pi^2$ Hz

5 Wolaita Sodo University · Final Exam 2014

A 1 kg particle undergoes SHM with amplitude 0.02 m and frequency 60 Hz. The maximum force on it is:

A) $288\pi^2$ N
B) $188\pi^2$ N
C) $144\pi^2$ N
D) $28.8\pi^2$ N

6 Wolaita Sodo University · Final Exam 2014

A 200 g block on a spring of force constant 5.00 N/m oscillates on a frictionless surface after being displaced 5.00 cm and released from rest. Find the period.

7 Wolaita Sodo University · Final Exam 2014

An 8.00 kg object moves with $x(t) = (5.00\ \text{m})\sin\left(\pi t + \frac{\pi}{6}\right)$. Find (a) the amplitude, frequency and period, (b) the maximum speed and acceleration, and (c) the total mechanical energy.

8 Wolaita Sodo University · Final Exam 2014

An object oscillates with $x(t) = 5.10\sin\left(1.2\pi t - \frac{\pi}{6}\right)$ m. Find (a) the amplitude, (b) the period, (c) the maximum acceleration, (d) the maximum velocity.

9 Model question (module activity)

A cut of beef thrown on a spring scale oscillates with period 0.500 s and amplitude 2.00 cm. Find (a) the frequency, (b) the maximum acceleration, (c) the maximum velocity, (d) the acceleration when the displacement is 1.00 cm.

10 Addis Ababa University · Final 2014

Which statement is true for the speed v and acceleration a of a particle in SHM?

A) When v is zero, a is zero
B) When v is maximum, a is maximum
C) When v is maximum, a is zero
D) a is zero whatever the value of v

11 Addis Ababa University · Final 2014

A 4 kg object on a spring oscillates with angular frequency 0.5 rad/s and amplitude 1.0 m. Find (a) the total energy, (b) the maximum speed, (c) the position where the speed is one quarter of its maximum.

12 Model question (module activity)

For a simple harmonic oscillator, which pair of vectors can NEVER point in the same direction?

A) Position and velocity
B) Velocity and acceleration
C) Position and acceleration
D) None of these

Answers & Explanations

1. **D**

There must be a constant velocity

In SHM the velocity changes continuously (maximum at the centre, zero at the ends). A, B and C are the conditions for SHM.

2. **B**

The acceleration is proportional to the displacement and opposite in direction

$a = -\omega^2 x$ defines SHM. The amplitude is constant, the period is independent of amplitude, and energy is conserved.

3. **C**

Frequency

Frequency f = cycles per second (Hz). The period is the time per cycle.

4. **C**

2π Hz

$v_{\max} = A\omega \Rightarrow \omega = 20\pi^2/5 = 4\pi^2$ rad/s → $f = \omega/2\pi = 2\pi$ Hz.

5. **A**

288π² N

$\omega = 2\pi(60) = 120\pi$ rad/s; $F_{\max} = m\omega^2 A = (1)(14\,400\pi^2)(0.02) = 288\pi^2$ N.

6. **1.26 s**

$T = 2\pi\sqrt{m/k} = 2\pi\sqrt{0.200/5.00} = 2\pi(0.200) = 1.26$ s (independent of the 5.00 cm amplitude).

7. **(a) 5.00 m, 0.500 Hz, 2.00 s; (b) 15.7 m/s, 49.3 m/s²; (c) 987 J**

(a) $\omega = \pi$ → $f = 0.500$ Hz, $T = 2.00$ s.
(b) $v_{\max} = 5\pi = 15.7$ m/s; $a_{\max} = 5\pi^2 = 49.3$ m/s².
(c) $E = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}(8.00)(5\pi)^2 = 100\pi^2 = 987$ J.

8. **(a) 5.10 m; (b) 1.67 s; (c) 72.4 m/s²; (d) 19.2 m/s**

$\omega = 1.2\pi$ rad/s → $T = 2\pi/\omega = 1.67$ s.
$a_{\max} = \omega^2 A = (1.2\pi)^2(5.10) = 72.4$ m/s²; $v_{\max} = \omega A = 19.2$ m/s.

9. **(a) 2.00 Hz; (b) 3.16 m/s²; (c) 0.251 m/s; (d) 1.58 m/s²**

$\omega = 2\pi/T = 4\pi = 12.6$ rad/s.
(b) $\omega^2 A = 158 \times 0.0200 = 3.16$ m/s². (c) $\omega A = 0.251$ m/s. (d) $|a| = \omega^2 x = 158 \times 0.0100 = 1.58$ m/s².

10. **C**

When v is maximum, a is zero

At the equilibrium position the speed is maximum and $a = -\omega^2 x = 0$. At the extremes $v = 0$ and a is maximum.

11. **(a) 0.5 J; (b) 0.5 m/s; (c) x = ±0.97 m**

(a) $E = \frac{1}{2} m\omega^2 A^2 = \frac{1}{2}(4)(0.25)(1.0) = 0.5$ J.
(b) $v_{\max} = \omega A = 0.5$ m/s.
(c) $\omega\sqrt{A^2 - x^2} = \frac{1}{4}\omega A \Rightarrow x = A\sqrt{15/16} = \pm 0.97$ m.

12. **C**

Position and acceleration

$\vec{a} = -\omega^2 \vec{x}$ always points opposite to the displacement. The other pairs point the same way during part of each cycle.

CHAPTER 5 · LESSON 5.2

The Simple Pendulum

AFTER THIS LESSON YOU CAN

- Derive and use the period of a simple pendulum
- Explain what the period does and does not depend on
- Use the pendulum to measure g
- Apply energy conservation to a swinging pendulum

5.2.1 Period of a simple pendulum

A **simple pendulum** is a small bob of mass m on a light string of length L. For **small angles** (less than about 15°), the restoring force $-mg\sin\theta \approx -mg\,x/L$ is proportional to the displacement, so the motion is SHM with $\omega = \sqrt{g/L}$:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

★ WHAT THE PERIOD DEPENDS ON

Depends on	Does NOT depend on
length L ($T \propto \sqrt{L}$)	the **mass** of the bob
gravity g ($T \propto 1/\sqrt{g}$)	the **amplitude** (for small swings)

- Double the length → period × $\sqrt{2}$; four times the length → period doubles.
- On the Moon ($g/6$) a pendulum swings $\sqrt{6} \approx 2.45$ times **slower**.
- A **seconds pendulum** ($T = 2.00$ s) has $L = g/\pi^2 \approx 0.993$ m.
- **Geophysics:** measuring T precisely gives $g = 4\pi^2 L/T^2$; small changes in g reveal oil or mineral deposits underground.

✎ WORKED EXAMPLE 1: MEASURING G

A 1.00 m pendulum makes 50 oscillations in 100.4 s. Find g.

$T = 100.4/50 = 2.008$ s → $g = \dfrac{4\pi^2 L}{T^2} = \dfrac{4\pi^2(1.00)}{2.008^2} = 9.79$ m/s².

✎ WORKED EXAMPLE 2: A ROCK ON A STRING (MODULE, CORRECTED)

A rock swings in a circle making 50 revolutions in 30 s. $f = 50/30 = 1.67$ Hz and $T = 1/f = 0.60$ s. (The printed module answer of 2 s is a slip.)

! PENDULUM TRAPS

- A heavier bob does **not** change the period.
- $T = 2\pi\sqrt{L/g}$ is valid only for **small** amplitudes.
- A pendulum clock taken up a mountain (smaller g) runs **slow** (longer period).

Exam Practice

5 QUESTIONS · LESSON 5.2

1 Wolaita Sodo University · Final Exam 2014

If the length of a simple pendulum is doubled, its period becomes:

A) twice the original period
B) four times the original period
C) half the original period
D) $\sqrt{2}$ times the original period

2 Model question

The bob of a simple pendulum is replaced by one with four times the mass. The period:

A) doubles
B) halves
C) quadruples
D) stays the same

3 Serway / OpenStax-style problem

A pendulum has a period of 2.0 s on Earth. On the Moon, where g is one sixth of Earth's, its period is about:

A) 4.9 s
B) 0.82 s
C) 0.33 s
D) 12 s

4 Model question

What length of pendulum has a period of exactly 1.00 s ($g = 9.80$ m/s²)?

5 Arba Minch University · Mid Exam

A pendulum of length L is released from rest at a height $L/4$ above its lowest point. Find its speed at the lowest point.

Answers & Explanations

1. D

$\sqrt{2}$ times the original period

$T \propto \sqrt{L}$, so doubling L multiplies T by $\sqrt{2} \approx 1.41$.

2. D

stays the same

$T = 2\pi\sqrt{L/g}$ contains no mass.

3. **A**

4.9 s

$T \propto 1/\sqrt{g}$: $T_{\text{Moon}} = 2.0\sqrt{6} = 4.9$ s.

4. **0.248 m**

$$L = \frac{gT^2}{4\pi^2} = \frac{9.80(1.00)^2}{39.5} = 0.248 \text{ m}.$$

5. **v = √(gL/2)**

Energy conservation: $mg\frac{L}{4} = \frac{1}{2}mv^2 \Rightarrow v^2 = \frac{gL}{2}$.

CHAPTER 5 · LESSON 5.3

Waves and Their Characteristics

AFTER THIS LESSON YOU CAN

- Define a wave and distinguish mechanical and electromagnetic waves
- Distinguish transverse and longitudinal waves
- Use v = fλ and the wave function
- Describe reflection, refraction, diffraction and interference

5.3.1 What is a wave?

D WAVE

A **disturbance** from equilibrium that **travels** through space, carrying **energy and momentum without transporting matter**. A leaf on a pond bobs up and down as ripples pass, but it is **not carried along**. A **pulse** is a single disturbance.

F WAVE VOCABULARY

Term	Meaning
Crest / trough	points of maximum displacement above / below equilibrium
Amplitude A	maximum displacement from equilibrium
Wavelength λ	distance between two consecutive points **in phase** (crest to crest)
Period T	time for the wave to travel one wavelength
Frequency f	oscillations per second, $f = 1/T$
Wave speed	$v = f\lambda = \dfrac{\lambda}{T}$

The speed is fixed by the **medium**; the frequency is fixed by the **source**. When a wave enters a new medium, f stays the same and λ changes.

5.3.2 Types of waves

D BY THE NEED FOR A MEDIUM

- **Mechanical waves** need a medium: water waves, **sound**, waves on a string, seismic waves. They need a source, a medium, and a way for parts of the medium to push on each other.
- **Electromagnetic (EM) waves** are produced by **accelerated charges** and can travel through **vacuum** at $c = 3.0 \times 10^8$ m/s: light, radio and TV waves, microwaves, X-rays, γ-rays.

D BY THE DIRECTION OF VIBRATION

- **Transverse:** particles vibrate **perpendicular** to the direction of travel: waves on a string, ripples on water, **all EM waves**.
- **Longitudinal:** particles vibrate **parallel** to the direction of travel, forming compressions and rarefactions: **sound**.

Travelling waves move through space; **standing waves** are trapped in an enclosure (a guitar string, an organ pipe).

F THE WAVE FUNCTION AND SOME SPEEDS

- Sinusoidal travelling wave: $y(x,t) = A\sin(kx - \omega t)$, with $k = \dfrac{2\pi}{\lambda}$ (wave number) and $\omega = 2\pi f$, so $v = \omega/k$.
- Wave on a string with tension F_T and mass per unit length μ: $v = \sqrt{F_T/\mu}$.
- Sound in air at 20 °C: $v \approx 343$ m/s (faster in warm air, much faster in water and steel).
- Standing waves on a string fixed at both ends: $\lambda_n = 2L/n$, $f_n = \dfrac{nv}{2L}$ $(n = 1, 2, 3, \ldots)$.

✎ WORKED EXAMPLE 1: RADIO AND SOUND

(a) An FM station broadcasts at 100 MHz. Find the wavelength. (b) A sound of 680 Hz travels at 340 m/s. Find its wavelength.

(a) $\lambda = c/f = 3.0 \times 10^8 / 1.00 \times 10^8 = 3.0$ m. (b) $\lambda = 340/680 = 0.50$ m.

5.3.3 Wave behaviour

F FOUR WAVE PHENOMENA

Phenomenon	What happens	Example
Reflection	a wave bounces back from a boundary	echo; a pulse on a string reflected from a wall (inverted at a fixed end)
Refraction	a change of **direction** as a wave passes into another medium, caused by a change in **speed**	ocean waves turning parallel to the beach; sound carrying far across a lake at night
Diffraction	waves **spread around obstacles** or through openings; strongest when λ is about the size of the gap	hearing someone around a corner; no perfectly sharp shadows
Interference	overlapping waves **add** (superposition): **constructive** (crest meets crest → larger) or **destructive** (crest meets trough → cancel)	noise-cancelling headphones; colours in soap films; **beats**
Beats: two sounds of slightly different frequencies give a loudness that rises and falls at $f_{\text{beat}} = \lvert f_1 - f_2 \rvert$ (used to tune instruments).		

! WAVE TRAPS

- Sound is a **mechanical, longitudinal** wave; it cannot travel through a vacuum.
- Water waves and sound are **not** EM waves.
- Waves transfer **energy**, not matter.
- In $v = f\lambda$, raising the frequency in the same medium **shortens** the wavelength; the speed is unchanged.

Exam Practice

8 QUESTIONS · LESSON 5.3

1 Wolaita Sodo University · Final Exam 2014

Water waves and sound waves are examples of electromagnetic waves.

True / False

2 Wolaita Sodo University · Final Exam 2014

X-rays, television waves and radio waves are examples of:

A) Mechanical waves
B) Longitudinal waves
C) Transverse mechanical waves
D) Electromagnetic waves

3 Wolaita Sodo University · Final Exam 2014

The change in direction of a wave passing from one medium to another, caused by a change in its speed, is:

A) Interference
B) Reflection
C) Diffraction
D) Refraction

4 Model question

In a longitudinal wave, the particles of the medium vibrate:

A) perpendicular to the direction of travel
B) parallel to the direction of travel
C) in circles
D) not at all

5 Model question

A wave has frequency 50 Hz and wavelength 4.0 m. Its speed is:

A) 200 m/s
B) 54 m/s
C) 12.5 m/s
D) 0.08 m/s

6 Serway / OpenStax-style problem

You can hear people talking around the corner of a building but cannot see them. This is mainly because:

A) sound travels faster than light
B) light cannot travel through air
C) sound is reflected but light is not
D) sound waves have wavelengths comparable to the doorway and diffract strongly; light waves have very short wavelengths

7 Serway / OpenStax-style problem

A travelling wave is described by $y = 0.050\sin(2.0\pi x - 40\pi t)$ in SI units. Find the amplitude, wavelength, frequency and speed.

8 Model question

Two tuning forks of 440 Hz and 444 Hz are sounded together. What do you hear?

Answers & Explanations

1. **False**

Both are **mechanical** waves that need a medium. EM waves (light, radio, X-rays) can travel through a vacuum.

2. **D**

Electromagnetic waves

They are all parts of the EM spectrum, travelling at c in vacuum.

3. **D**

Refraction

That is the definition of refraction.

4. **B**

parallel to the direction of travel

Longitudinal (e.g. sound) waves have particle motion along the direction of propagation, forming compressions and rarefactions.

5. **A**

200 m/s

$v = f\lambda = 50 \times 4.0 = 200$ m/s.

6. **D**

sound waves have wavelengths comparable to the doorway and diffract strongly; light waves have very short wavelengths

Diffraction is strongest when the wavelength is comparable to the obstacle; sound wavelengths are about 1 m, light wavelengths about 5×10^{-7} m.

7. **A = 0.050 m; λ = 1.0 m; f = 20 Hz; v = 20 m/s**

$k = 2.0\pi \Rightarrow \lambda = 2\pi/k = 1.0$ m. $\omega = 40\pi \Rightarrow f = 20$ Hz. $v = f\lambda = 20$ m/s (or ω/k).

8. **A tone of about 442 Hz whose loudness rises and falls 4 times per second**

Beat frequency $= |444 - 440| = 4$ Hz.

CHAPTER 5 · LESSON 5.4

Resonance and the Doppler Effect

AFTER THIS LESSON YOU CAN

- Explain natural frequency and resonance with examples
- Explain the Doppler effect qualitatively
- Apply the Doppler formula with the correct signs
- Recognise applications of both

5.4.1 Resonance

D RESONANCE

The **large increase in amplitude** of an oscillating system when it is driven by a periodic force whose **frequency equals (or is very close to) its natural frequency**. The frequency at which this happens is the **resonant frequency**.

E RESONANCE AROUND US

- **Pushing a swing:** small pushes timed at the swing's natural frequency build a large amplitude.
- **Musical instruments:** guitar bodies, flute and organ air columns resonate to amplify particular notes.
- A singer shattering a wine glass; tuning a **radio** (the circuit resonates with one station's frequency); **MRI** scanners (nuclear magnetic resonance).
- **Dangers:** soldiers break step on bridges; the 1940 Tacoma Narrows Bridge collapse; engineers design buildings to avoid resonance with earthquakes and wind.

5.4.2 The Doppler effect

D DOPPLER EFFECT

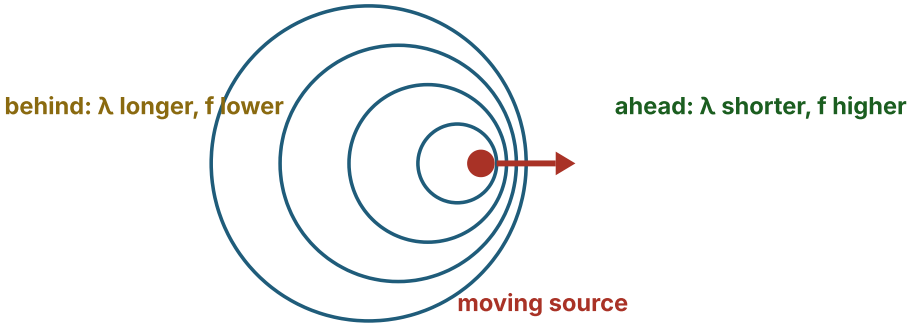
The **apparent change in frequency** heard by an observer when the **source and observer move relative to each other**. Approaching → **higher** frequency (higher pitch); receding → **lower** frequency. **The source's actual frequency does not change**; the waves are bunched up in front of a moving source and stretched out behind it.

$$f_o = f_s \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

v = speed of sound, v_o = observer's speed, v_s = source's speed.

★ SIGN RULE

Use the **upper signs** ($+v_o$ on top, $-v_s$ below) for motion **toward** the other; the **lower signs** for motion **away**. Check: approaching must give a **higher** frequency.



Wavefronts crowd together in front of a moving source

✎ WORKED EXAMPLE 1: AMBULANCE SIREN

An ambulance siren emits 800 Hz and moves at 30 m/s; sound travels at 340 m/s. What frequency does a stationary listener hear as it approaches and as it recedes?

Approaching: $f_o = 800\left(\frac{340}{340-30}\right) = 877$ Hz. Receding: $f_o = 800\left(\frac{340}{340+30}\right) = 735$ Hz.

E APPLICATIONS OF THE DOPPLER EFFECT

Police **radar** speed guns; **Doppler ultrasound** of blood flow in medicine; weather radar; bats' echolocation; astronomers use the **red shift** of light from galaxies to show the universe is expanding.

Exam Practice

5 QUESTIONS · LESSON 5.4

1 Wolaita Sodo University · Final Exam 2014

Fill in the blank: "The _____ effect is observed whenever there is relative motion between a wave source and an observer."

2 Model question (module)

As a police car with its siren on drives toward you and then past you, the pitch you hear:

A) is high, then low
B) is low, then high
C) stays the same
D) rises continuously

3 Model question (module)

In the Doppler effect, the frequency emitted by the source actually changes.

True / False

4 Model question (module)

A large increase in amplitude when a system is driven at its natural frequency is called:

A) damping
B) refraction
C) diffraction
D) resonance

5 Serway / OpenStax-style problem

A train whistle of 500 Hz approaches a stationary observer at 25 m/s (speed of sound 340 m/s). What frequency is heard? What if the observer instead moves at 25 m/s toward a stationary whistle?

Answers & Explanations

1. **Doppler**

The observed frequency changes with relative motion: higher when approaching, lower when receding.

2. **A**

is high, then low

Approaching → compressed waves → higher frequency; receding → stretched waves → lower frequency.

3. **False**

The source frequency is unchanged; only the frequency **received** by the observer changes because of the relative motion.

4. **D**

resonance

That is the definition of resonance (e.g. pushing a swing in time with its natural swing).

5. **540 Hz; 537 Hz**

Moving source: $f_o = 500\left(\frac{340}{340-25}\right) = 540$ Hz.

Moving observer: $f_o = 500\left(\frac{340+25}{340}\right) = 537$ Hz. (Close, but not identical.)

CHAPTER 5 · LESSON 5.5

Image Formation by Mirrors and Thin Lenses

AFTER THIS LESSON YOU CAN

- Describe images in plane mirrors
- Distinguish converging and diverging lenses and mirrors
- Use the thin-lens/mirror equation and magnification with correct signs
- Predict image position and nature with ray diagrams

5.5.1 Plane mirrors

Reflection obeys the **law of reflection**: angle of incidence = angle of reflection (both measured from the normal). A plane mirror forms an image that is:

- **virtual** (the rays only **appear** to come from behind the mirror),
- **upright** and the **same size** as the object,
- **as far behind** the mirror as the object is in front,
- **laterally inverted** (left and right swapped, as in "AMBULANCE" written backwards on vehicles).

Two plane mirrors at an angle θ give $n = \frac{360^\circ}{\theta} - 1$ images (e.g. 90° → 3 images).

5.5.2 Lenses and curved mirrors

★ CONVERGING (POSITIVE F)

- **Convex lens** (thicker at the centre) and **concave mirror**.
- Parallel rays meet at a **real focus** F.
- Can form **real** or **virtual** images.

! DIVERGING (NEGATIVE F)

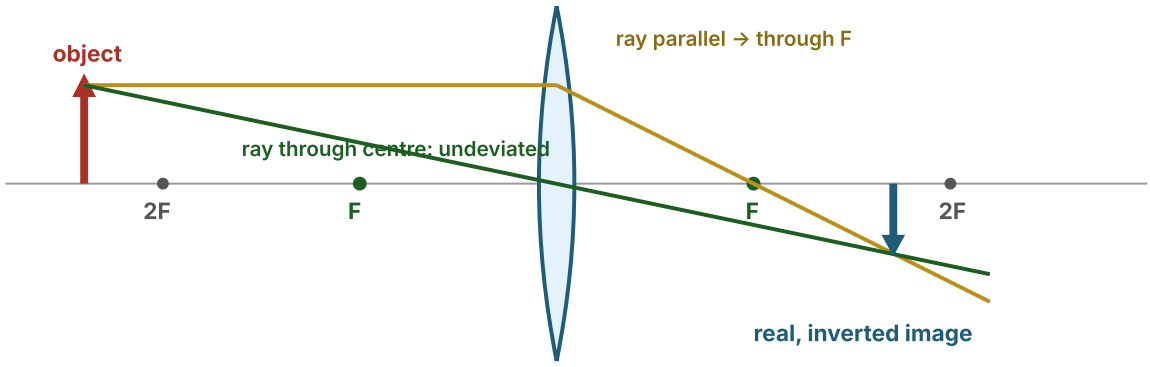
- **Concave lens** (thinner at the centre) and **convex mirror**.
- Parallel rays spread out as if from a **virtual focus**.
- Always form **virtual, upright, diminished** images of real objects.

The **focal length** f is the distance from the lens (or mirror) to the focal point. For a spherical mirror, $f = R/2$. The **power** of a lens is $P = 1/f$ (in **dioptres**, D, with f in metres).

ƒ THIN-LENS AND MIRROR EQUATION

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \qquad m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

Sign convention ("real is positive"): $d_o > 0$ for a real object; $d_i > 0$ for a **real** image, $d_i < 0$ for a **virtual** image; $f > 0$ converging, $f < 0$ diverging. $m < 0$: **inverted**; $m > 0$: **upright**; $|m| > 1$: enlarged.
(The module writes s_o, s_i for d_o, d_i and $m = h_i/h_o = s_i/s_o$ in magnitude.)



Convex lens, object beyond 2F: the image is real, inverted and diminished, between F and 2F

IMAGES FORMED BY A CONVEX (CONVERGING) LENS

Object position	Image position	Nature	Size
At infinity	at F	real, inverted	point (highly diminished)
Beyond 2F	between F and 2F	real, inverted	diminished
At 2F	at 2F	real, inverted	**same size**
Between 2F and F	beyond 2F	real, inverted	enlarged
At F	at infinity	(no image formed)	(rays emerge parallel)
Between F and lens	same side as object	**virtual, upright**	**enlarged** (magnifying glass)

Concave (diverging) lens: for any real object the image is **virtual, upright and diminished**, between F and the lens.

WORKED EXAMPLE 1: CONVEX LENS, TWO CASES

A convex lens has $f = 10$ cm. Find the image for an object at (a) 30 cm, (b) 5.0 cm.

(a) $\frac{1}{d_i} = \frac{1}{10} - \frac{1}{30} = \frac{2}{30} \Rightarrow d_i = +15$ cm; $m = -15/30 = -0.5$: **real, inverted, half size**.

(b) $\frac{1}{d_i} = \frac{1}{10} - \frac{1}{5} = -\frac{1}{10} \Rightarrow d_i = -10$ cm; $m = -(-10)/5 = +2$: **virtual, upright, twice the size** (a magnifier).

WORKED EXAMPLE 2: CONCAVE LENS

A concave lens has $f = -20$ cm; the object is 30 cm away.
$\frac{1}{d_i} = -\frac{1}{20} - \frac{1}{30} = -\frac{5}{60} \Rightarrow d_i = -12$ cm; $m = +12/30 = +0.40$: virtual, upright, diminished.

E APPLICATIONS OF IMAGE FORMATION

Cameras and the **human eye** (convex lens → real, inverted, diminished image on the film or retina); **magnifying glasses**; **projectors** (object between F and 2F → enlarged real image); **microscopes and telescopes** (two lenses); **spectacles** (convex for long sight, concave for short sight); **car side mirrors** (convex: wide view, diminished); **shaving and dental mirrors** (concave: enlarged).

! OPTICS TRAPS

- A **concave lens** never forms a real image of a real object; a **concave mirror** can (it converges).
- A plane mirror's image is always **virtual** and the same size.
- Object **at F** of a convex lens: **no image** (rays parallel).
- Be consistent with signs: a negative image distance means **virtual**.

Exam Practice

7 QUESTIONS · LESSON 5.5

1 Wolaita Sodo University · Final Exam 2014

For a concave lens, the images of real objects are always:

A) erect, same size and virtual
B) erect, diminished and virtual
C) real, inverted and enlarged
D) real, diminished and inverted

2 Model question (module)

The image formed by a plane mirror is:

A) real and inverted
B) virtual, upright and the same size
C) virtual and enlarged
D) real and the same size

3 Model question (module)

An object is placed at 2F of a convex lens. The image is:

A) at F, real and diminished
B) virtual and enlarged
C) beyond 2F, real and enlarged
D) at 2F, real, inverted and the same size

4 Model question (module)

Where must an object be placed in front of a convex lens to obtain a virtual, upright, enlarged image?

A) Beyond 2F
B) At 2F
C) Between F and 2F
D) Between F and the lens

5 Serway / OpenStax-style problem

An object 4.0 cm tall stands 15 cm from a convex lens of focal length 10 cm. Find the image distance, magnification and image height, and describe the image.

6 Serway / OpenStax-style problem

A concave mirror has a radius of curvature of 40 cm. An object is placed 10 cm in front of it. Locate the image.

7 Model question

A lens has a power of +2.5 D. It is:

A) a converging lens with f = 40 cm
B) a diverging lens with f = 40 cm
C) a converging lens with f = 2.5 m
D) a diverging lens with f = 25 cm

Answers & Explanations

1. **B**

erect, diminished and virtual

A diverging lens always forms a virtual, upright, diminished image between the focal point and the lens.

2. **B**

virtual, upright and the same size

A plane mirror forms a virtual, upright image of the same size, as far behind the mirror as the object is in front.

3. **D**

at 2F, real, inverted and the same size

$d_o = 2f \Rightarrow d_i = 2f$ and $m = -1$.

4. **D**

Between F and the lens

Inside the focal length the rays diverge after the lens; their backward extensions form a virtual, enlarged image (magnifying glass).

5. **d_i = +30 cm; m = −2.0; h_i = −8.0 cm: real, inverted, enlarged**

$\frac{1}{d_i} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$; $m = -30/15 = -2.0$; $h_i = mh_o = -8.0$ cm.

6. **d_i = −20 cm (virtual, upright, magnified × 2)**

$f = R/2 = 20$ cm. $\frac{1}{d_i} = \frac{1}{20} - \frac{1}{10} = -\frac{1}{20} \Rightarrow d_i = -20$ cm (behind the mirror); $m = -d_i/d_o = +2$.

7. **A**

a converging lens with f = 40 cm

$f = 1/P = 1/2.5 = 0.40$ m; positive power means converging.

CHAPTER 5 · QUICK REVISION

Oscillations, Waves and Optics — at a glance

CHAPTER 5 FORMULA SHEET

Topic	Formulas
Oscillation basics	$f = 1/T;\ \omega = 2\pi f$
SHM	$a = -\omega^2 x;\ x = A\sin(\omega t + \phi);\ v_{\max} = \omega A;\ a_{\max} = \omega^2 A$
Spring	$\omega = \sqrt{k/m};\ T = 2\pi\sqrt{m/k};\ E = \frac{1}{2}kA^2$
Pendulum	$T = 2\pi\sqrt{L/g}$ (small angles)
Waves	$v = f\lambda;\ y = A\sin(kx - \omega t);\ k = 2\pi/\lambda;\ v = \sqrt{F_T/\mu};\ f_{\text{beat}} = \lvert f_1 - f_2\rvert$
Doppler	$f_o = f_s\dfrac{v \pm v_o}{v \mp v_s}$ (upper signs: toward)
Optics	$\dfrac{1}{d_o} + \dfrac{1}{d_i} = \dfrac{1}{f};\ m = -\dfrac{d_i}{d_o} = \dfrac{h_i}{h_o};\ f_{\text{mirror}} = R/2;\ P = 1/f$

CONVEX LENS IN ONE SENTENCE

"As the object walks in from infinity to F, the real image walks out from F to infinity and grows; inside F the image jumps to the object's side, becomes virtual, upright and big."